Timothy Wang

Frontend Engineer · React / TypeScript / Next.js · Accessibility & Testing

Sydney, Australia · timothy.zehao.wang@gmail.com

linkedin.com/in/timothy-z-wang · github.com/timzw961 · timzwang.dev

Frontend engineer specialising in React, TypeScript and Next.js, most recently building booking and checkout journeys at Qantas Hotels. Strong focus on accessibility, automated testing and production reliability, from cutting ~25,000 daily checkout errors to under 100 to bringing E2E feedback from 40+ minutes to under 10.

EXPERIENCE

Self-employed *Jun 2026 – Present*

Software Engineer, AI Evaluation

- Design frontend engineering tasks and evaluate AI coding agents' implementations against requirements, identifying functional bugs, missed edge cases and code quality issues.
- Write evaluation rubrics and structured feedback assessing agent reasoning and solution quality.

Qantas Group *Aug 2023 – Mar 2026*

Senior Frontend Engineer | **Qantas Hotels** *Apr 2025 – Mar 2026*

- Built React/TypeScript features across Qantas Hotels' search, property details, booking and checkout journeys, supporting ~A$2M in daily transaction value, including a travel insurance cross-sell on booking confirmation.
- Investigated ~25,000 daily quote failures using Splunk, tracing them to missing checkout validation. Implemented quote and date validation with a recovery flow, reducing errors to fewer than 100 per day and preventing unintended booking-date changes.
- Independently overhauled Cypress E2E tests for critical booking journeys, restoring skipped coverage, stabilising flaky tests and adding payment and quote-failure scenarios. Reduced test runtime in Buildkite from 40+ minutes to under 10 minutes.
- Co-designed a multi-layer accessibility testing strategy with QE, implementing the developer-facing layers (eslint-plugin-jsx-a11y, axe-core, jest-axe) and fixing keyboard and semantic issues found through VoiceOver testing, materially strengthening WCAG 2.1 AA conformance ahead of specialist audit.
- Implemented GA4 e-commerce tracking across the booking funnel and migrated legacy analytics to a shared data-helper layer.
- Delivered Points + Pay improvements, including lowering the minimum points threshold to 2,000 and fixing points-slider, deposit-payment and post-login checkout autofill issues.
- Implemented Optimizely location-based targeting for double/triple-points promotions, enabling 10+ production campaigns to launch through configuration rather than code releases.
- Mentored a technology graduate through code reviews and pairing, helping scope a 500+ file Enzyme-to-RTL migration into a team-wide effort. Personally migrated 250+ test files across the Qantas and Jetstar Hotels codebases using a reviewed Gemini-assisted workflow.

Technology Graduate | **Rotational Program** *Aug 2023 – Apr 2025*

- Rotated across Qantas Money frontend engineering, Core Platforms Technology Enablement and Group Cyber Automation.
- **Qantas Money:** Delivered Next.js features and GA4 instrumentation across the new Qantas Pay website ahead of the Qantas Pay app launch, tracking the migration of existing Qantas Travel Money customers and keeping user journeys measurable across native app and webview pages.
- **Core Platforms:** Investigated a 30s/60s timeout mismatch between two services that triggered duplicate travel insurance redemptions, leading to a backend duplicate check that eliminated repeat charges.

Digital Transformation Graduate | **HLB Mann Judd** *Feb 2023 – Aug 2023*

- Automated reporting and client engagement workflows with Power Automate, removing weeks of manual processing per cycle and contributing to six-figure annual savings.

SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Redux, HTML5, CSS3/Sass, Tailwind, Vite, Storybook

Testing & Accessibility: Jest, React Testing Library, Cypress, Playwright, jest-axe, axe-core, Pa11y, WCAG 2.1 AA, ARIA, semantic HTML

Platforms: SSR/SSG, BFF patterns, Node.js, REST, GraphQL, Optimizely, Contentful, GA4

Tooling: Buildkite, GitHub Actions, Git, Splunk, Postman

AI Tooling: Claude Code, Gemini, OpenAI Codex

PROJECTS

Rally Reserve | Next.js 16, React 19, TypeScript, PostgreSQL, Prisma, Auth.js, Zod, Jest + RTL, Playwright Live app

- Built a full-stack court booking platform for racquet sports with multi-tenant venues, per-venue timezones, recurring bookings and a time-credit loyalty program.
- Prevented double-bookings under concurrent requests with row-level locking and overlap checks inside a single PostgreSQL transaction, and stopped loyalty credit being spent twice with guarded conditional updates, both verified by Playwright tests firing simultaneous requests.
- Wrote 450+ automated tests across Jest/React Testing Library and Playwright E2E against an isolated PostgreSQL database, with tests, linting and type checks running on every pull request in GitHub Actions.

EDUCATION

Bachelor of Computer Science (Software Engineering) | University of Wollongong *2019 – 2022*
High Distinction Average · Dean's Merit List (2020–2022)